

Applicability of Current Theories of Flow in Leaky Aquifers

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Abstract. The applicability of current theories of flow in leaky aquifers has been investigated using new analytic and numeric solutions. In general it has been customary to use one or both of the following assumptions: (1) storage in the aquitards is negligible, and (2) drawdowns in the unpumped aquifers remain zero. At small values of time the first assumption amounts to neglecting the effects of leakage completely. Errors introduced in this way can become significant when the β factor of Hantush exceeds 0.01. It appears that β may exceed 0.1 in many field situations. At large values of time, neglecting drawdowns in the unpumped aquifers can also lead to serious errors unless the transmissibilities of these aquifers are significantly greater than that of the pumped aquifer. In addition present methods of analyzing leaky aquifers usually rely only on drawdown data from the pumped aquifer. In our opinion such an approach is not sufficient to characterize a leaky system; current methods of analyzing field data from leaky aquifers need to be improved.

NOTATIONS

b_i ,	thickness of i th aquifer, L ;
b_j' ,	thickness of j th aquitard, L ;
$J_0(x)$,	zero order Bessel function of the first kind;
K_i ,	permeability of i th aquifer, L/T ;
K_j' ,	permeability of j th aquitard, L/T ;
Q_i ,	pumping rate from i th aquifer, L^3/T ;
r ,	radial distance from pumping well, L ;
r/B_{ij} ,	$r\sqrt{K_j'/K_i b_j'}$;
$S_{s,i}$,	specific storage of i th aquifer, L^{-1} ;
$S_{s,j}'$,	specific storage of j th aquitard, L^{-1} ;
s_i ,	drawdown in i th aquifer, L ;
s_j' ,	drawdown in j th aquitard, L ;
s_D ,	dimensionless drawdown ($4\pi T_s/Q_i$);
T_i ,	transmissibility of i th aquifer ($K_i b_i$), L^2/T ;
t ,	pumping time, T ;
$t_{D,i}$,	dimensionless time for pumped (i th) aquifer ($K_i t/S_{s,i} r^2$);
z ,	vertical distance above bottom of aquitard 1, L ;
β_{ij} ,	$r/4b_i\sqrt{K_j'S_{s,j}'/K_i S_{s,i}}$.

INTRODUCTION

When water is being withdrawn from the subsurface, the question often arises as to how much of the production comes from the aquifer in which the well is completed and how much comes from beds lying above or below. An aquifer that receives a significant inflow from adjacent beds while being pumped is called a 'leaky' aquifer. In reality a leaky aquifer is often just one part of a multiple aquifer system, i.e., a succession of aquifers separated by varying thicknesses of relatively low permeabil-

ity aquitards and/or aquicludes. If these intervening beds are sufficiently low in permeability, they act like no-flow boundaries, and the behavior of any aquifer that is being pumped should follow the Theis solution [Neuman and Witherspoon, 1968].

However, as permeability increases, leakage becomes appreciable and flow is not restricted to the pumped aquifer alone. The additional water is derived primarily from storage in the aquitards and the adjoining unpumped aquifers. Since flow is not restricted to the pumped aquifer alone, the leaky aquifer must be viewed as part of a complex multiple aquifer system where the drawdown in each layer depends on the flow behavior of the entire system. Therefore one cannot develop a complete understanding of flow in the pumped aquifer without having analyzed the system as a whole.

Since a rigorous approach to flow in multiple aquifer systems involves boundary conditions that make the problem intractable analytically, it has been customary to simplify the mathematics by assuming that flow is essentially horizontal in the aquifers and vertical in the aquitards. The validity of this assumption has recently been investigated by Neuman and Witherspoon [1969a] for the case of two aquifers and one aquitard. It was found that when the permeabilities of the aquifers are more than two orders of magnitude greater

than that of the aquitard the errors introduced by this assumption are usually less than 5%. These errors increase with time, decrease with radial distance from the pumping well, and are smallest in the pumped aquifer and greatest in the aquitard. Since the permeability contrast between aquifer and aquitard is often greater than three orders of magnitude, it would appear that the above assumptions can be used safely.

In dealing with steady state problems the assumption of vertical flow in the aquitards implies that the rate of leakage into the pumped aquifer is proportional to the potential drop across the leaky aquitard. An approach to steady flow in leaky aquifers based on this assumption was first introduced into the literature by *De Glee* [1930] and later was used by *Steggeventz and Van Ness* [1939]. Additional work along these lines by *Glebov* [1940], *Myatiev* [1946, 1947], and *Girinsky* [1947] has been summarized by *Polubarinova-Kochina* [1962] and *Aravin and Numerov* [1965]. Only in one case described by *Polubarinova-Kochina* [1962, p. 385] has the drawdown been allowed to vary in more than one aquifer at a time.

Jacob [1946] used the same approach to develop a partial differential equation for nonsteady flow in a leaky aquifer. However, it is important to recognize that under transient conditions the assumption that leakage is proportional to the potential drop across the aquitard is tantamount to ignoring the storage capacity of the aquitard. In addition Jacob assumed that head in the unpumped aquifer remained constant. A similar approach has been used by *Polubarinova-Kochina* [1962, p. 546] to investigate the problem of seepage from a ditch into an aquifer underlain by an aquitard.

The above assumptions have greatly facilitated mathematical treatment of the problem. As a result between the years 1949-1960 *Hantush* [1949, 1956, 1957, 1959] and *Hantush and Jacob* [1954, 1955a, 1955b, 1960] have used this approach to develop a large number of solutions to various problems involving flow in aquifers with vertical leakage. One of these solutions [*Hantush and Jacob*, 1955b] describes nonsteady radial flow to a well that completely penetrates an infinite leaky aquifer and discharges at a constant rate. This solution is of

particular interest in the present work because it has been extensively tabulated [*Hantush*, 1956], and the resulting type curves are being widely used by ground-water hydrologists to evaluate the properties and potential yield of leaky aquifers [*Ferris et al.*, 1962; *Walton*, 1960, 1962; *Slater*, 1963; *De Wiest*, 1965; *Narasimhan*, 1968]. The results of this solution are usually presented in terms of a dimensionless parameter

$$r/B = r \sqrt{\frac{K'}{Kbb'}} \quad (1)$$

which we note is a function of the permeability contrast between the pumped aquifer and an adjacent aquitard. We shall refer to this as the '*r/B* solution.'

Subsequent to the *r/B* solution *Hantush* [1960a] published a modified version of the differential equations for nonsteady flow in a leaky system in which for the first time consideration was given to the effect of storage in the aquitards. Assuming that drawdown in the unpumped aquifers remains zero, *Hantush* was able to develop asymptotic solutions for the pumped aquifer that apply only at small and large values of time. We shall demonstrate later that *Hantush's* solutions are quite good over a broader time span than he has indicated. His solution for small values of time has been extensively tabulated [*Hantush*, 1960b] in terms of a dimensionless parameter

$$\beta = \frac{r}{4b} \sqrt{\frac{K'S_s'}{KS_s}} \quad (2)$$

which we see differs from *r/B* in that the effects of storage are now included. We shall refer to this as the '*β* solution.'

To our knowledge *Hantush's* modified approach has not been used in subsequent work on leaky aquifers. Between 1961-1967 *Hantush* [1962, 1964, 1967a, 1967b] and *De Wiest* [1961, 1963] analyzed various problems involving flow in leaky aquifers, but in all of these investigations storage in the aquitard was consistently neglected. Only in one analysis [*Hantush*, 1967b] has drawdown in the unpumped aquifer been allowed to vary with time.

Recently we have investigated the problem of flow in a system consisting of two aquifers separated by an aquitard [*Neuman and Witherspoon*, 1968].

spoon, 1969a, 1969b]. In contrast to previous work we considered the effects of storage in the aquitard and allowed drawdown in the unpumped aquifer to vary with time. Our solutions have been evaluated numerically [Neuman and Witherspoon, 1969a] and verified by the finite element method [Javandel and Witherspoon, 1968b; Witherspoon et al., 1968].

The β solution corresponds to a special case of this two aquifer problem wherein drawdown in the unpumped aquifer is assumed to remain zero, and the r/B solution corresponds to another special case where it is further assumed that storage in the aquitard can be neglected. To investigate the applicability of the β and r/B solutions as well as the underlying assumptions that were used in obtaining these solutions we must first introduce a complete solution for a two aquifer system with zero drawdown in the unpumped aquifer and consider the effects of storage in the aquitard.

SOLUTION FOR TWO AQUIFER SYSTEM WITH ZERO DRAWDOWN IN UNPUMPED AQUIFER

Consider a system composed of two aquifers and one aquitard (Figure 1). A well of infinitesimal diameter completely penetrates one of the aquifers and discharges at constant rate Q_1 . Each layer is homogeneous, isotropic, horizontal, and of infinite radial extent. No-flow boundaries are assumed for the upper and lower limits of the complete system, and it is further assumed that the aquifers and aquitard remain water-saturated at all times.

In the particular case where drawdown in the unpumped aquifer remains zero ($s_2 = 0$), the above problem becomes equivalent to Case 1 in Hantush's [1960a] 'Modification of the Theory of Leaky Aquifers.' His asymptotic solution for small values of time, which is the β solution, can be written

$$s_1(r, t) = \frac{Q_1}{4\pi T_1} \int_{1/(4tD_1)}^{\infty} \exp(-y) \cdot \operatorname{erfc}\left(\frac{\beta_{11}}{\sqrt{y(4tD_1y - 1)}}\right) \frac{dy}{y} \quad (3)$$

where Hantush's criterion for the validity of this solution can be expressed as

$$t_{D_1} \leq \frac{1.6\beta_{11}^2}{(r/B_{11})^4} \quad (4)$$

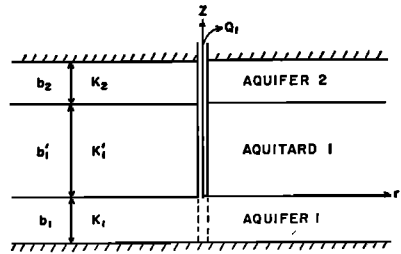


Fig. 1. Schematic diagram of two aquifer system.

Hantush's asymptotic solution for large values of time can be written

$$s_1(r, t) = \frac{Q_1}{4\pi T_1} \cdot \int_{1/(4tD_1)}^{\infty} \exp\left[-y - \frac{(r/B_{11})^2}{4y}\right] \frac{dy}{y} \quad (5)$$

where

$$\delta_1 = 1 + \frac{16\beta_{11}^2}{3(r/B_{11})^2} \quad (6)$$

and his criterion for the validity of this solution can be expressed as

$$t_{D_1} \geq \frac{80\beta_{11}^2}{(r/B_{11})^4} \quad (7)$$

It is seen by virtue of the expressions given for δ_1 and t_{D_1} in equations 4, 6, and 7 that (3) and (5) are functions of both β_{11} and r/B_{11} . In addition it is evident from (4) and (7) that for given values of β_{11} and r/B_{11} these solutions cover the entire time domain except for an interval whose span is less than two log cycles.

As mentioned above, a complete solution for this problem for all values of time has recently been developed by Neuman and Witherspoon [1969a, 1969b]. This solution may be written in terms of β_{11} and r/B_{11} as

$$s_1(r, t) = \frac{Q_1}{4\pi T_1} \cdot 2 \int_0^{\infty} (1 - e^{-y^2 t_{D_1}}) J_0(\omega(y)) \frac{dy}{y} \quad (8)$$

where

$$t_{D_1} = t_{D_1}(r/B_{11})^4 / (16\beta_{11}^2)$$

$$\omega^2(y) = \frac{(r/B_{11})^4}{16\beta_{11}^2} y^2 - (r/B_{11})^2 y \cot y$$

and $J_0[\omega(y)]$ must be set to zero when $\omega^s(y) < 0$. A solution for the more general case when the pumped aquifer is enclosed between two leaky aquitards has also been developed [Neuman and Witherspoon, 1969a].

Equation 8 has been evaluated numerically, and the results for $\beta_{11} = 0.01, 0.1$, and 1.0 are shown in Figures 2, 3, and 4, respectively. In each figure we see an envelope from which a family of r/B_{11} curves extend. The position of the envelope depends on the magnitude of β_{11} , and as this parameter increases in magnitude one sees that deviations from the Theis solution also increase. As the r/B_{11} parameter increases, the curves diverge from the envelope at earlier and earlier values of dimensionless time. These curves reach steady state values for dimensionless drawdown that are identical with those obtained by Hantush [1956].

Each curve on Figures 2, 3, and 4 has been divided by a set of parentheses into three sections. The section to the left of the parentheses corresponds to Hantush's [1960a] definition of small values of time (equation 4) from his β solution. Our results as obtained from equation 8 are identical with his β solution (given by equation 3) over these same time intervals.

The section to the right of the parentheses on each curve corresponds to Hantush's solution for large values of time (equation 7). Again our results as obtained from equation 8 are identical with those evaluated by Hantush [1960a] from equation 5.

The sections on each curve of Figures 2, 3, and 4 that are enclosed by parentheses therefore represent the time intervals for which Hantush [1960a] stated his asymptotic solutions should not be used. Our results indicate that his criteria for the validity of his solutions are on the conservative side and could be relaxed somewhat.

EVALUATING THE ASSUMPTION THAT STORAGE IN THE AQUITARD MAY BE NEGLECTED

As indicated earlier Hantush and Jacob [1955b] have solved the above problem of flow in a leaky aquifer by assuming that the storage capacity of the aquitard can be neglected. Their solution for all values of time can be expressed as

$$s_1(r, t) = \frac{Q_1}{4\pi T_1} \cdot \int_{1/(4tD_1)}^{\infty} \exp \left[-y - \frac{(r/B_{11})^2}{4y} \right] \frac{dy}{y} \quad (9)$$

and has been referred to here as the r/B solution. One may note that if storage is to be neglected, then $\beta_{11} = 0$ in (6), and equations 5 and 9 are identical.

Anyone familiar with the literature on leaky aquifers will immediately recognize that the family of curves shown in Figure 2 for $\beta = 0.01$ is almost identical with that of the r/B solution. The only difference is that the envelope of curves instead of coinciding with the Theis

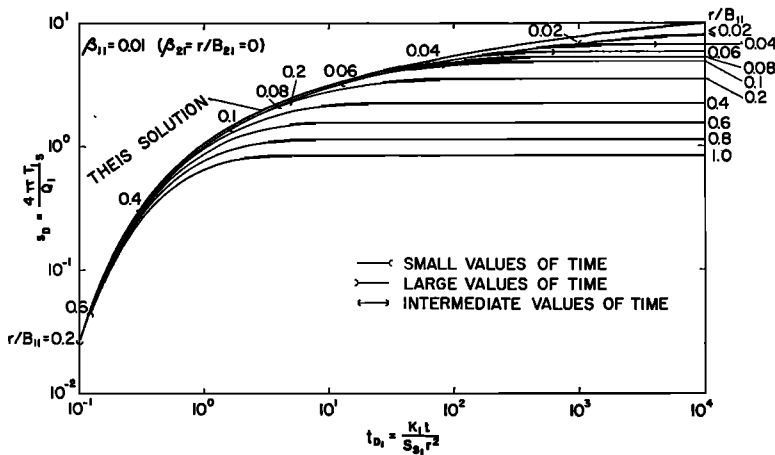


Fig. 2. Dimensionless drawdown in pumped aquifer when drawdown in unpumped aquifer is zero for $\beta_{11} = 0.01$.

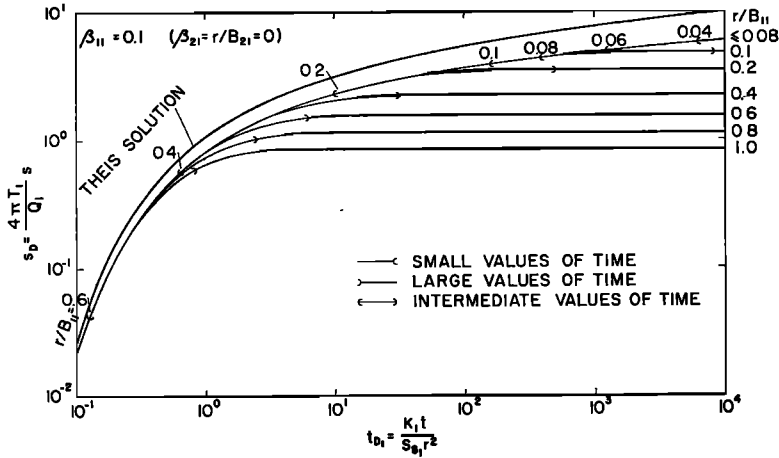


Fig. 3. Dimensionless drawdown in pumped aquifer when drawdown in unpumped aquifer is zero for $\beta_{11} = 0.1$.

solution slowly diverges from it as time increases.

Thus if one is analyzing field data at large values of time for a system with $\beta_{11} = 0.01$, ignoring the effects of storage in the aquiclude will only introduce a slight error as long as r/B_{11} is also small (i.e., $\lesssim 0.01$). From a practical standpoint, however, it would appear that neglecting storage in the aquitard should not affect the solution for the pumped aquifer as long as $\beta_{11} \leq 0.01$.

An examination of Figures 3 and 4 shows that as β_{11} increases, the r/B solution becomes less and less representative of the actual behavior

in the pumped aquifer. The errors involved in the r/B solution become significant when β_{11} reaches 0.1 (Figure 3), and they are large when $\beta_{11} = 1.0$ (Figure 4).

The nature of these errors for large values of β_{11} can be better understood by superposing the r/B solution on the $\beta_{11} = 1.0$ solution of Figure 4 as shown in Figure 5. One sees immediately that the errors involved in the r/B solution increase as the magnitude of r/B_{11} decreases. However, from the convergence of the curves for the two solutions at large values of time, it is also apparent that these errors decrease with time and disappear altogether for those values of t_D given by (7).

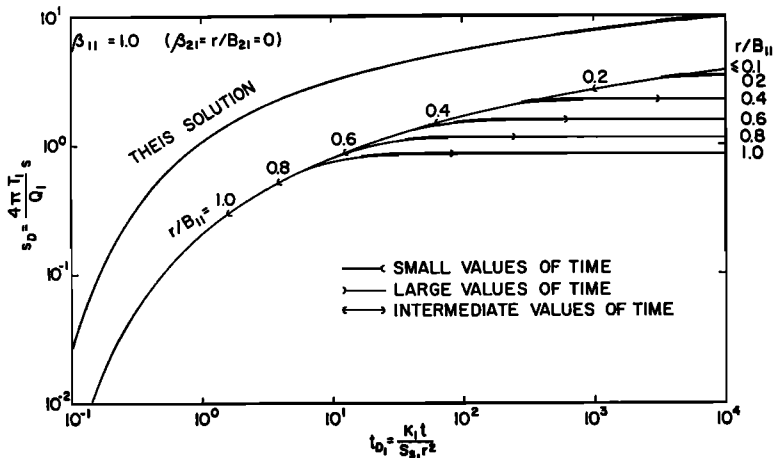


Fig. 4. Dimensionless drawdown in pumped aquifer when drawdown in unpumped aquifer is zero for $\beta_{11} = 1.0$.

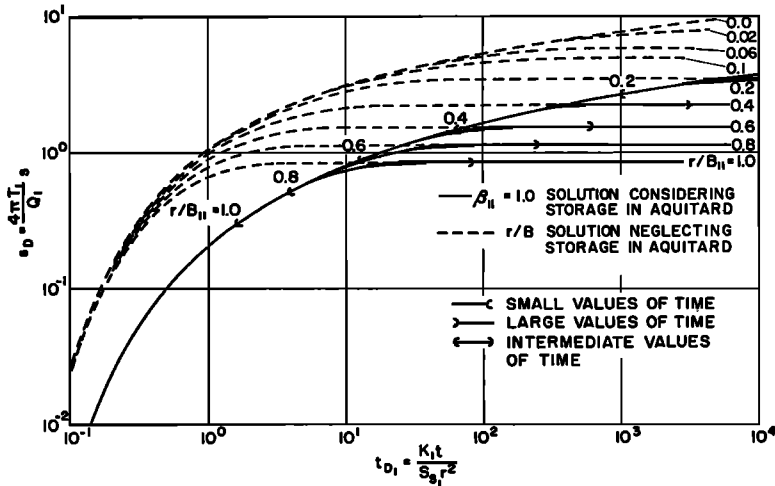


Fig. 5. Comparison of solution for pumped aquifer when $\beta_{11} = 1.0$ with r/B solution.

We therefore conclude that with large values of β_{11} and zero drawdown in the unpumped aquifer, the r/B solution is subject to significant errors whenever $t_{D1} \leq 80\beta_{11}^2/(\tau/B_{11})^4$.

For the particular case of $\beta_{11} = 1.0$ as shown on Figure 5, this means that the entire non-steady state period of pumping cannot be analyzed using the r/B solution. In other words the r/B solution would only be applicable to the steady state after drawdowns have become constant. It should be kept in mind that the steady state regions on all the curves of Figures 2 through 5 are the result of the initial assumption that drawdown remains zero in the unpumped aquifer.

There is a simple physical explanation for the errors in the r/B solution when β_{11} is large. At small values of time the disturbance created by withdrawing water has not yet significantly affected the unpumped aquifer. Most of the early leakage is derived from the aquitard, and the amount depends on the specific storage of this part of the system, i.e., on the magnitude of β_{11} . At this stage, disregarding storage in the aquitard is equivalent to neglecting leakage altogether. Thus the resulting curves for the r/B solution fail to show the true effects of leakage at early time.

As time increases, more and more leakage is being contributed by the unpumped aquifer, and the relative amount of water that comes from storage in the aquitard diminishes. By

the time steady state is reached and drawdowns are constant, all of the leakage is supplied by the unpumped aquifer. The aquitard merely acts as a conduit for flow from one aquifer to another. The storage capacity of the aquitard has no influence on the behavior of the system, and therefore the r/B solution is applicable.

The assumption that storage in the aquitard may be neglected obviously fails completely if one is interested in the transient behavior of the aquitard itself. When drawdown in the unpumped aquifer remains zero, a solution has been obtained for the aquitard [Neuman and Witherspoon, 1969a, 1969b] which can be written

$$s_1'(r, z, t) = \frac{Q_1}{4\pi T_1} \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi z}{b_1'} \cdot \int_0^{\infty} \left[e^{-n^2 \pi^2 i_{D1}} - 1 - \frac{e^{-n^2 \pi^2 i_{D1}} - e^{-y^2 i_{D1}}}{1 - y^2/(n^2 \pi^2)} \right] \cdot J_0[\omega(y)] \frac{dy}{y} \quad (10)$$

where i_{D1} and $\omega(y)$ are as defined in (8) and $J_0[\omega(y)]$ must be set to zero when $\omega^2(y) < 0$. Equation 10 has been evaluated for selected values of β_{11} and τ/B_{11} , and the results demonstrate that s_1' does not become linear in z until the system has reached steady state (Figures 4-2 through 4-9, Neuman and Witherspoon [1969a]).

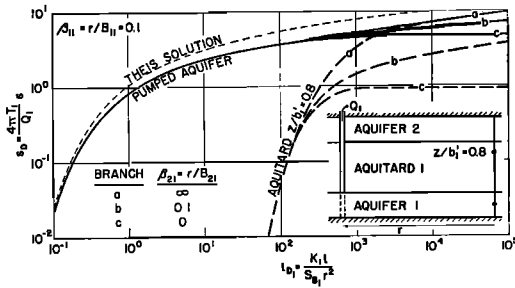


Fig. 6. Effect of unpumped aquifer on dimensionless drawdown in pumped aquifer and in aquitard at $z/b_1 = 0.8$.

EVALUATING THE ASSUMPTION THAT DRAWDOWN IN THE UNPUMPED AQUIFER MAY BE NEGLECTED

As mentioned earlier we have recently obtained a complete solution to the problem of flow in a two aquifer system for the case when drawdown in the unpumped aquifer is not necessarily zero [Neuman and Witherspoon, 1969a, 1969b]. The results must be expressed in terms of four parameters: (a) β_{11} and r/B_{11} with reference to the pumped aquifer and (b) β_{21} and r/B_{21} with reference to the unpumped aquifer. For the pumped aquifer the solution may be expressed as

$$s_1(r, t) = \frac{Q_1}{4\pi T_1} \int_0^\infty (1 - e^{-v^* i_{D_1}}) \cdot \{ [1 + G(y)] J_0[\omega_1(y)] + [1 - G(y)] J_0[\omega_2(y)] \} \frac{dy}{y} \quad (11)$$

For the aquitard the solution is

$$s_1'(r, z, t) = \frac{Q_1}{4\pi T_1} \frac{2}{\pi} \sum_{n=1}^\infty \frac{1}{n} \sin \frac{n\pi z}{b_1'} \cdot \int_0^\infty \left[1 - e^{-n^2 \pi^2 i_{D_1}} + \frac{e^{-n^2 \pi^2 i_{D_1}} - e^{-v^* i_{D_1}}}{1 - y^2/(n^2 \pi^2)} \right] \cdot \left\{ \left[\frac{2(r/B_{21})^2 (-1)^n y}{F(y) \sin y} - G(y) - 1 \right] J_0[\omega_1(y)] - \left[\frac{2(r/B_{21})^2 (-1)^n y}{F(y) \sin y} - G(y) + 1 \right] \cdot J_0[\omega_2(y)] \right\} \frac{dy}{y} \quad (12)$$

and for the unpumped aquifer

$$s_2(r, t) = \frac{Q_1}{4\pi T_1} \int_0^\infty (1 - e^{-v^* i_{D_1}}) \frac{2(r/B_{21})^2}{F(y)} \cdot \{ J_0[\omega_1(y)] - J_0[\omega_2(y)] \} \frac{dy}{\sin y} \quad (13)$$

where

$$G(y) = M(y)/F(y)$$

$$\omega_1^2(y) = \frac{1}{2}[N(y) + F(y)]$$

$$\omega_2^2(y) = \frac{1}{2}[N(y) - F(y)]$$

$$F^2(y) = M^2(y) + \left[\frac{2(r/B_{11})(r/B_{21})y}{\sin y} \right]^2$$

$$M(y) = \left[\frac{(r/B_{11})^4}{\beta_{11}^2} - \frac{(r/B_{21})^4}{\beta_{21}^2} \right] \frac{y^2}{16}$$

$$- \left[\left(\frac{r}{B_{11}} \right)^2 - \left(\frac{r}{B_{21}} \right)^2 \right] y \cot y$$

$$N(y) = \left[\frac{(r/B_{11})^4}{\beta_{11}^2} + \frac{(r/B_{21})^4}{\beta_{21}^2} \right] \frac{y^2}{16}$$

$$- \left[\left(\frac{r}{B_{11}} \right)^2 + \left(\frac{r}{B_{21}} \right)^2 \right] y \cot y$$

As in the case of (8), $J_0[\omega_1(y)]$ must be set to zero when $\omega_1^2(y) < 0$, and the same is true of $J_0[\omega_2(y)]$.

These solutions have been evaluated for selected values of the four controlling parameters, and some of the results for the particular case when $\beta_{11} = r/B_{11} = 0.1$ are shown in Figure 6. These results have also been checked by using the finite element method [Javandel and Witherspoon, 1968b; Witherspoon et al., 1968].

As might be expected the results in the pumped aquifer are independent of β_{21} and r/B_{21} at small values of time. In general we found that the transient behavior of the pumped aquifer and the aquitard was not affected by conditions in the unpumped aquifer as long as equation 4 is satisfied. However, transient effects in the unpumped aquifer are dependent on β_{21} and r/B_{21} as well as on β_{11} and r/B_{11} at all values of time. Thus the assumption that drawdown in the unpumped aquifer may be neglected is valid at small values of time everywhere in the system except in the unpumped aquifer itself.

At larger values of time when $t_{D_1} > 1.6\beta_{11}^2/(r/B_{11})^4$, the behavior of the unpumped aquifer

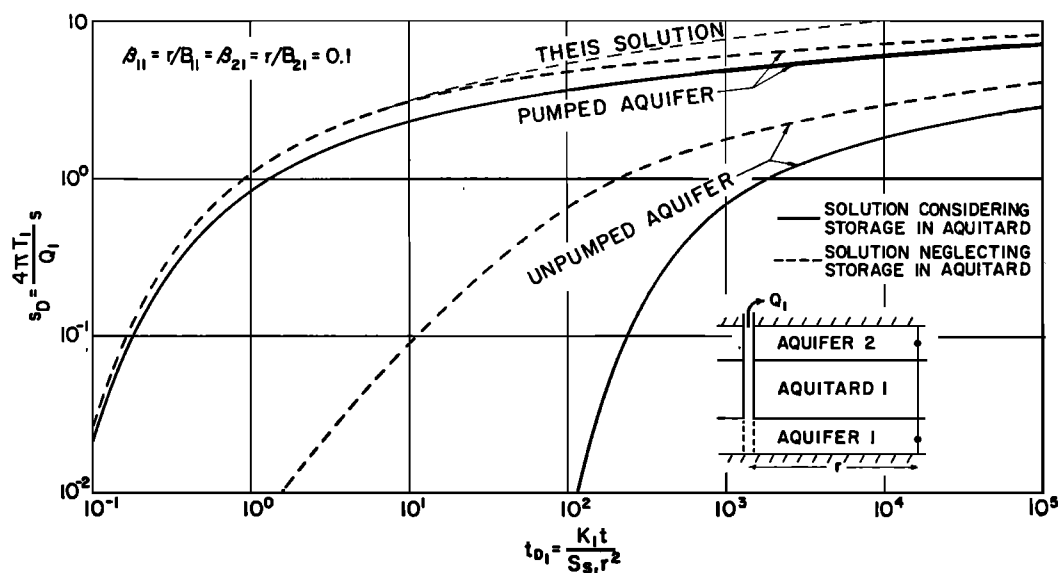


Fig. 7. Effect of neglecting storage in aquitard on dimensionless drawdown in pumped and unpumped aquifers (evaluated from equations 11, 13, 14, and 15).

may have a significant effect on drawdown in other parts of the system. For example, reference to Figure 6 will show that the results for the pumped aquifer that are uniquely defined for a given β_{11} and r/B_{11} at small values of time become a family of curves at large values of time, depending on the values of β_{21} and r/B_{21} .

A family with three branches is shown for the drawdown curve of the pumped aquifer in Figure 6. The lower branch corresponds to the special case previously discussed in connection with Figures 2-4, where it was assumed that there is no drawdown in the unpumped aquifer. A necessary condition for zero drawdown in the unpumped aquifer is that its transmissibility be infinitely large. Thus the lower branch of the curves for the pumped aquifer on Figure 6 corresponds to the special case when $T_2 = \infty$, which means $\beta_{21} = r/B_{21} = 0$.

When β_{21} and r/B_{21} are greater than zero, the assumption of no drawdown in the unpumped aquifer will no longer be valid. For example the middle branch of the drawdown curve for the pumped aquifer on Figure 6 represents another special case where the hydraulic properties of both aquifers are identical, i.e. $\beta_{11} = \beta_{21}$ and $r/B_{11} = r/B_{21}$. The upper branch represents the limiting case when $\beta_{21} = r/B_{21} = \infty$. This latter case implies that the per-

meability of the unpumped aquifer is zero ($K_2 = 0$) and means that a no-flow boundary exists at the interface between the aquiclude and the unpumped aquifer. This corresponds to Hantush's [1960a] case 2 in his modified treatment of leaky aquifers. It is clear from Figure 6 that when the transmissibility of the unpumped aquifer is not infinitely large, as would usually be the case in the field, drawdown in the pumped aquifer does not reach steady state.

The assumption of zero drawdown in the unpumped aquifer can therefore lead to errors that will depend primarily on the magnitude of β_{11} and r/B_{11} . As is indicated on Figure 6, where $\beta_{11} = r/B_{11} = 0.1$, these errors increase with time as the ratios β_{21}/β_{11} and $(r/B_{21})/(r/B_{11})$ increase, i.e., as the transmissibility and storage capacity of the unpumped aquifer become less in relation to those of the pumped aquifer.

However, as the values of β_{11} and r/B_{11} decrease, the errors introduced by assuming zero drawdown in the unpumped aquifer will decrease for given ratios of T_2/T_1 and S_{s2}/S_{s1} . Theoretically these errors will not disappear completely unless the unpumped aquifer is replaced by a ponded body of water whose head remains constant. From a practical standpoint these errors

can probably be neglected in analyzing drawdowns in the pumped aquifer when both β_{11} and $r/B_{11} < 0.01$ or when the ratios T_2/T_1 and S_{u1}/S_{u2} are sufficiently large. Our present data suggest that β_{21} and r/B_{21} should be about ten times smaller than β_{11} and r/B_{11} , respectively, in order for the errors to be negligible. This theory needs further investigation.

If we now turn our attention briefly to the aquitard, one would anticipate that the properties of the unpumped aquifer will have a much more profound effect on drawdown in the aquitard than we have just seen for the pumped aquifer. To illustrate this point we have included results on Figure 6 determined at only one location in the aquitard ($z/b_1' = 0.8$). Radial distance from the pumping well also becomes important, and these aquitard results are for a distance of more than twice the combined thicknesses of pumped aquifer and aquitard ($b_1 + b_1'$).

The aquitard results shown on Figure 6 for the limiting case of $\beta_{21} = r/B_{21} = \infty$ reveal an interesting result. It will be noted that dimensionless drawdown in the aquitard coincides with that of the pumped aquifer at large values of time. This means that in this limiting case of a no-flow boundary on one side of the aquitard, flow in all parts of the system away from the immediate vicinity of the pumping well becomes radial at large values of time. *Javandel and Witherspoon* [1968a] report the same effect has been noted for a well that partially penetrates a two-layered aquifer regardless of the permeability contrast.

Thus since it has been customary to assume vertical flow in the aquitard in developing solutions for leaky aquifers, the special case of a no-flow boundary presents a problem because the direction of flow is essentially vertical only at early time and then slowly changes to become radial at large values of time, especially as distance from the pumping well increases. *Hantush's* [1960a] asymptotic solution at large values of time for case 2 in his modified approach to leaky aquifers does not take this into consideration. However, in checking his results for the pumped aquifer, we found that our solution, which in this case was obtained by the finite element method, is in good agreement with his, indicating that direction of flow

in the aquitard at this stage does not affect the result in the pumped aquifer.

As mentioned earlier, *Hantush* [1967b] has also analyzed this two aquifer case and developed solutions for drawdown in both the pumped and the unpumped aquifer. However, he assumed that storage in the intervening aquitard could be neglected. It is therefore of interest to see how our results, which include the effects of storage in the aquitard, compare with his. If we choose $T_1 = T_2$ (e.g., $r/B_{11} = r/B_{21}$), then *Hantush's* equations 17 and 18 for the special case of equal diffusivities in both aquifers ($K_1/S_{u1} = K_2/S_{u2}$) may be written

$$s_1(r, t) = \frac{Q_1}{8\pi T_1} \int_{1/(4tD_1)}^{\infty} \left\{ \exp(-y) + \exp \left[-y - \frac{2(r/B_{11})^2}{4y} \right] \right\} \frac{dy}{y} \quad (14)$$

$$s_2(r, t) = \frac{Q_1}{8\pi T_1} \int_{1/(4tD_1)}^{\infty} \left\{ \exp(-y) - \exp \left[-y - \frac{2(r/B_{11})^2}{4y} \right] \right\} \frac{dy}{y} \quad (15)$$

Figure 7 shows a comparison of results for the pumped and unpumped aquifers as obtained using our solutions (equations 11 and 13) and as obtained using *Hantush's* solutions (equations 14 and 15). In both cases we arbitrarily chose $\beta_{11} = \beta_{21} = r/B_{11} = r/B_{21} = 0.1$. One may note that in the pumped aquifer our results differ somewhat from those of *Hantush*. His solution lies along the Theis solution at early time and then diverges from it as time increases so as to lie between our solution and the Theis curve. If we had chosen the case for $\beta_{11} = \beta_{21} = 1.0$, these differences would be considerably greater.

It should be recalled that when we had zero drawdown in the unpumped aquifer, errors due to neglecting storage in the aquitard did not disappear until steady state was reached. Now we see from Figure 7 that when there is no steady state these errors occur at all values of time.

For the unpumped aquifer, however, it is seen that there is a large difference between our results and those of *Hantush*. His solution gives an earlier response and a much greater drawdown than ours. This of course is to be

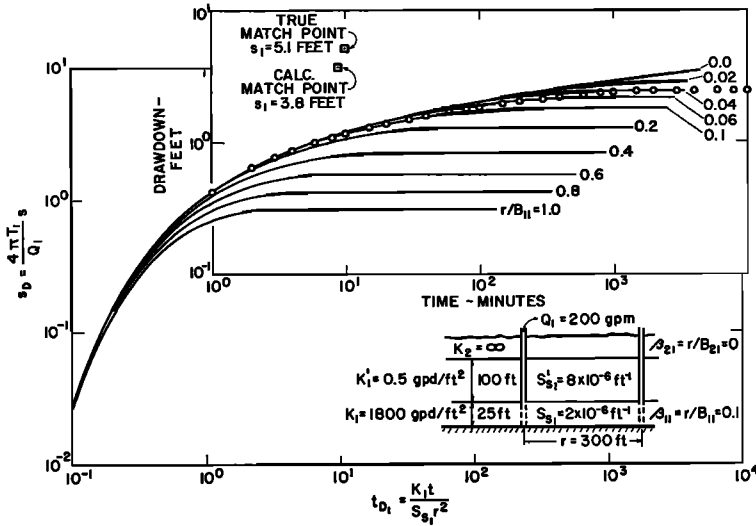


Fig. 8. Comparison of hypothetical field data with r/B solution when drawdown in unpumped aquifer is zero.

expected because in our system the aquitard contributes water from storage and therefore acts as a buffer between the two aquifers. As might be expected the differences between our two solutions for the unpumped aquifer will be even greater if $\beta_{11} = \beta_{21} = 1.0$.

This naturally raises the question whether the magnitude of the storage coefficient for an aquitard is so small that it can be neglected. This question is of course difficult to answer because of the general lack of data on the hydrologic characteristics of aquitards. We have found one comprehensive report containing a number of laboratory measurements on core samples from several different aquitards of Central California [Johnson *et al.*, 1965]. The materials range from sandy silts to clays. Permeabilities range from 10^{-6} to over 10^{-3} gpd/ft², and specific storage ranges from 3×10^{-6} to 5×10^{-4} ft⁻¹. In other words these data indicate that the specific storage of unconsolidated aquitard materials is at least as large as that of most confined aquifer sands and gravels, if not significantly greater. Under these circumstances it is not difficult to show that β may easily reach 0.1 and at times may exceed 1.0. If this is the case, our analysis indicates that, in general, storage in the aquitard must be taken into consideration when evaluating leaky systems.

ERRORS INTRODUCED BY USING r/B SOLUTION TO ANALYZE FIELD DATA

The following two examples of hypothetical field data will be used to illustrate the kind of errors that can arise when one uses the r/B solution to analyze the results of a pumping test.

For the first example we shall assume a two aquifer system (Figure 8) where $b_1 = b_2 = 25$ feet, $b_1' = 100$ feet, $K_1 = 1800$ gpd/ft² and $S_{s1} = 2 \times 10^{-6}$ ft⁻¹ for the aquifer being pumped, and $K_1' = 0.5$ gpd/ft² and $S_{s1}' = 8 \times 10^{-6}$ ft⁻¹ for the overlying aquitard. We shall further assume that $T_2 \gg T_1$ such that $\beta_{21} \approx r/B_{21} \approx 0$, which means that drawdown in the unpumped aquifer is negligible.

If an observation well is completed in aquifer 1 at a distance of $r = 300$ feet from the pumping well, the values of β_{11} and r/B_{11} become

$$\begin{aligned} \beta_{11} &= \frac{r}{4b_1} \sqrt{\frac{K_1' S_{s1}'}{K_1 S_{s1}}} \\ &= \frac{300}{(4)(25)} \sqrt{\frac{(0.5)(8 \times 10^{-6})}{(1800)(2 \times 10^{-6})}} = 0.1 \\ \frac{r}{B_{11}} &= r \sqrt{\frac{K_1'}{K_1 b_1 b_1'}} \\ &= 300 \sqrt{\frac{0.5}{(1800)(25)(100)}} = 0.1 \end{aligned}$$

Under these conditions a pumping test is performed with $Q_1 = 200$ gpm, and the drawdown data at the observation well would of course follow the curve for $\beta_{21} = r/B_{21} = 0.0$ as shown on Figure 6. This fact, however, is unknown to the analyst, and in using the r/B solution he would obtain the best match with $r/B = 0.04$ (Figure 8). Using the indicated match point at $s_D = 10$, where $s_1 = 3.8$ feet, he would then obtain

$$K_1 = \frac{114.6 Q_1 s_D}{b_1 s_1} = \frac{(114.6)(200)(10)}{(25)(3.8)} = 2410 \text{ gpd/ft}^2$$

$$K_1' = \frac{(r/B_{11})^2 K_1 b_1 b_1'}{r^2} = \frac{(0.04)^2 (2410)(25)(100)}{(300)^2} = 0.11 \text{ gpd/ft}^2$$

Thus we see that in using the r/B solution for this situation the calculated permeability is too high for the aquifer ($\sim 33\%$) and almost five times too low for the aquitard. The calculated permeability contrast would be $K_1/K_1' = 2410/0.11 = 21,900$, whereas the true value is only 3600.

If a second observation well were available at $r = 3000$ feet, then drawdown data would follow the curve for $\beta_{21} = r/B_{21} = 1.0$ shown on Figure 4. If the analyst again used the r/B solution in the same manner as above, he would probably obtain the best match with $r/B = 0.3$. At the same match point of $s_D = 10$, where $s_1 = 1.5$ feet, he would then obtain

$$K_1 = \frac{(114.6)(200)(10)}{(25)(1.5)} = 6110 \text{ gpd/ft}^2$$

$$K_1' = \frac{(0.3)^2 (6110)(25)(100)}{(3000)^2} = 0.15 \text{ gpd/ft}^2$$

From this second set of results we see that the calculated permeability is three times too high for the aquifer and three times too low for the aquitard. The calculated permeability contrast is worse than before, $K_1/K_1' = 6110/0.15 = 40,700$, or over ten times the true value. This illustrates how the errors increase as β_{21} and r/B_{21} increase. The analyst could be badly misled from the results of this analysis by

(a) overpredicting the productivity of the aquifer, (b) underestimating the leakage contribution from the aquitard, and (c) concluding that the aquifer is radially inhomogeneous when the apparent increase in permeability with distance is due to a misinterpretation of the effects of leakage.

For the second example we shall use the same two aquifer system as before, except that now we shall assume that the hydraulic properties of both aquifers are identical, i.e., $K_1 = K_2 = 1800$ gpd/ft² and $S_{21} = S_{22} = 2 \times 10^{-4}$ ft⁻¹. If the aquitard properties are the same as before, then an observation well at $r = 300$ feet would mean $\beta_{11} = \beta_{21} = r/B_{11} = r/B_{21} = 0.1$.

Under these conditions a pumping test is performed with $Q_1 = 200$ gpm, and the drawdown data would now follow the curve for $\beta_{21} = r/B_{21} = 0.1$ shown on Figure 6. This fact is again unknown to the analyst, and he might choose to match the field results with the r/B solution as indicated on Figure 9. If he does so, then at $s_D = 10$, $s_1 = 3.6$, and the permeability result for the aquifer would be $K_1 = 2540$ gpd/ft², or about 40% too high.

This result might be acceptable, but a more serious problem now arises in that one will have difficulty deciding what value of r/B to choose. The analyst who expects the drawdown data to flatten may be tempted to choose an r/B value at the end of the test period where he presumes the flattening is about to appear. However, the longer the test runs, the lower the value of r/B that will be selected, and the more the result will be in error.

Another alternative for the analyst is to ignore the early drawdown data and shift the plot of field data so as to force a match with the Theis solution. This would yield even larger errors in K_1 and also lead to the erroneous conclusion that there is no leakage.

CONCLUSIONS

It is our conclusion that present methods of analyzing field data from leaky aquifers need to be improved. These methods are based on two assumptions that we believe are not generally applicable. The assumption that storage in the aquitard is negligible can lead to significant errors when β of the pumped aquifer is greater than 0.01. It appears that large values

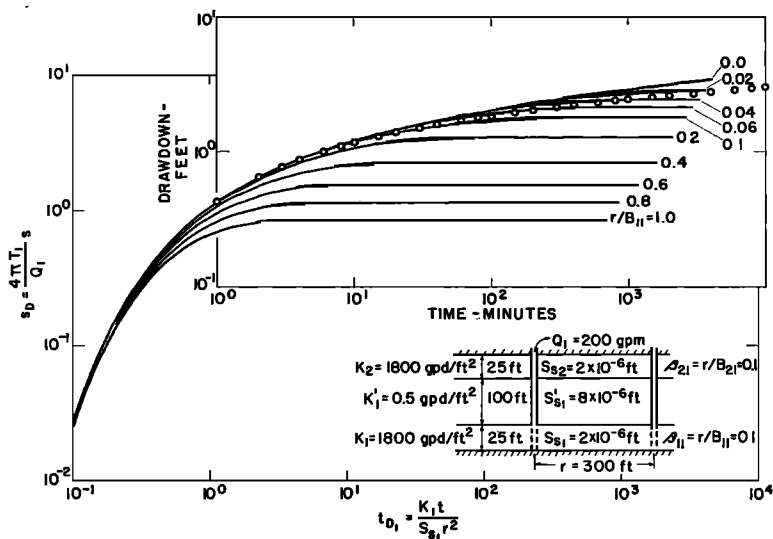


Fig. 9. Comparison of hypothetical field data with r/B solution when properties of aquifers are identical.

of β of the order of 0.1 to 1.0 are certainly possible in many field situations.

The second assumption of zero drawdown in the unpumped aquifer can also lead to significant errors at large values of time. These errors cannot be neglected unless the transmissibility of the unpumped aquifer is significantly greater than that of the pumped aquifer. Since the r/B solution relies on both of these assumptions, one must be cautious in using it to analyze field data.

Our entire discussion has been restricted to the case of a rather simple two aquifer system. Obviously the problems of evaluating field data are further complicated by the fact that leakage may occur not only from above but also from below. This will tend to restrict further the applicability of the above two assumptions.

It seems to us that relying entirely on drawdown data obtained from the pumped aquifer is not sufficient to characterize a leaky system. Our theory [Neuman and Witherspoon, 1969a, 1969b] suggests that by using drawdown data from both aquifers and aquitards one should be able to develop improved methods of analysis. We are currently attempting to develop such methods.

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